

Diploma Level Web Development Notes (Basic to Advanced)

Chapter 1: Introduction to Web Development

Web development is the process of creating websites and web applications. It consists of Frontend Development, Backend Development, Database Management, Web Security and Deployment. A website can be static or dynamic. Static websites show fixed content while dynamic websites interact with databases and users.

Chapter 2: HTML Fundamentals

HTML stands for HyperText Markup Language. It is the foundation of every website. Important topics include document structure, headings, paragraphs, links, images, lists, tables, forms, multimedia, semantic tags and HTML5 features.

Chapter 3: CSS Fundamentals

CSS stands for Cascading Style Sheets. CSS is used to design and style websites. Topics include selectors, colors, fonts, backgrounds, box model, positioning, display properties, flexbox, grid system, animations and media queries.

Chapter 4: Responsive Web Design

Responsive design allows websites to adapt to desktops, tablets and mobile devices. Important concepts include viewport, flexible layouts, responsive images and media queries.

Chapter 5: JavaScript Basics

JavaScript adds interactivity to web pages. Topics include variables, operators, conditions, loops, arrays, objects, functions, events and DOM manipulation.

Chapter 6: Advanced JavaScript

Advanced JavaScript includes ES6 features, arrow functions, promises, async/await, modules, APIs and error handling.

Chapter 7: Bootstrap Framework

Bootstrap provides pre-designed UI components. Important topics include containers, rows, columns, buttons, forms, cards, navbars and responsive grids.

Chapter 8: PHP Programming

PHP is a server-side scripting language. Topics include variables, arrays, functions, forms, sessions, cookies, file handling and database connectivity.

Chapter 9: Python Flask Framework

Flask is a lightweight Python framework. Topics include routing, templates, forms, sessions, Jinja2, blueprints and deployment.

Chapter 10: Database Management

Databases store information permanently. SQL commands include CREATE, INSERT, SELECT, UPDATE and DELETE. Database normalization improves efficiency.

Chapter 11: MySQL and TiDB

MySQL and TiDB are relational databases. Important concepts include tables, primary keys, foreign keys, joins, indexing and transactions.

Chapter 12: Authentication and Authorization

Authentication verifies identity while authorization controls access. Techniques include password hashing, sessions, JWT tokens and OTP verification.

Chapter 13: REST APIs

REST APIs allow communication between applications. HTTP methods include GET, POST, PUT and DELETE. JSON is the most common data format.

Chapter 14: AJAX and Fetch API

AJAX allows data loading without refreshing pages. The Fetch API is commonly used to communicate with backend servers.

Chapter 15: Git and GitHub

Git tracks code changes while GitHub hosts repositories online. Developers use branching, commits, pull requests and collaboration workflows.

Chapter 16: Web Security

Important threats include SQL Injection, Cross-Site Scripting (XSS), Cross-Site Request Forgery (CSRF), session hijacking and weak passwords.

Chapter 17: Docker

Docker packages applications with dependencies. Containers improve portability, consistency and deployment efficiency.

Chapter 18: Cloud Computing

Cloud computing provides computing resources over the internet. Popular providers include AWS, Azure and Google Cloud.

Chapter 19: CI/CD

Continuous Integration and Continuous Deployment automate testing, building and deployment processes.

Chapter 20: Mini Projects

Student Management System, Library Management System, StudyHub, Attendance System, Online Quiz Platform, Notes Sharing Website, E-learning Platform and AI Chat Applications.

Chapter 21: Viva Questions

1. What is HTML? 2. Difference between GET and POST? 3. What is CSS? 4. What is JavaScript? 5. What is Bootstrap? 6. What is Flask? 7. What is a Database? 8. What is API? 9. What is Authentication? 10. Difference between SQL and NoSQL?

Chapter 22: Interview Questions

Explain DOM. What is Responsive Design? Explain Sessions and Cookies. Difference between Authentication and Authorization. Explain REST APIs. What is Docker? Explain Git Workflow. What is MVC Architecture?

Chapter 23: StudyHub Project Guide

For your StudyHub project: 1. User Registration and Login 2. Admin Panel 3. Notes Upload 4. Video Lectures 5. Categories and Search 6. Comments and Feedback 7. Progress Tracking 8. Mobile Responsive Design 9. Email Verification 10. Deployment on Cloud.

Conclusion

These diploma-level notes provide a complete overview of web development from beginner concepts to advanced deployment and security topics. They are useful for semester exams, viva preparation, interviews and project development.